

Inferring the environment from a few context samples instead of predefined labels makes latent dynamics **identifiable** and **generalizable to unseen environments**.

Meta-iLaD jointly identifies latent state z_t , environment c , and dynamics F by meta-learning across environments — the **first** identifiability of c & F under a general $F(z; c)$.

1 The Problem & Setup

High-dimensional time series — **video, motion, health** — are driven by **low-dimensional latent dynamics**: a latent state z_t evolved by a dynamics function and emitted through g .

$$x_t = g(z_t) + \epsilon, \quad z_t = F(z_{t-1}; c), \quad 0 \leq t \leq T$$

F = Neural ODE

$$\frac{dz_t}{dt} = f_g(z_t; c)$$

Recover them *from data alone* — i.e. **identifiability** at three levels:

 z_t

state

 c

environment

 F

dynamics fn.

2 The Identifiability Gap

Modelled **unconditionally**, the system is **not identifiable**:

$$p_\Theta(x_{0:T}) = \int \int p(c) p(z_0) \prod_{t=1}^T p_\Theta(z_t | z_{t-1}, c) \prod_{t=0}^T p_g(x_t | z_t) dz_{0:T} dc$$

Prior work conditions F on **predefined labels** u — **no OOD, useless without labels**. Identifiability of c & F themselves stays **barely studied**, yet decides prediction.

3 Key Idea

Replace labels with **k-shot context samples** D^s that share the environment; infer c from a **conditional prior** — a feed-forward meta-learner:

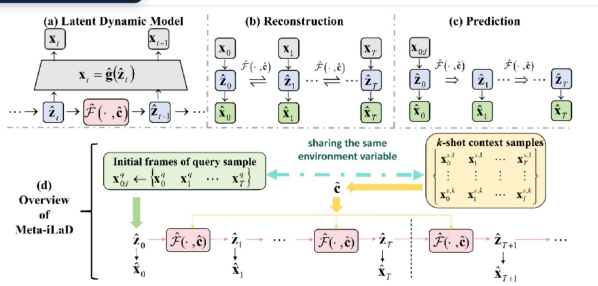
$$p_\zeta(c | D^s) = \prod_{i=1}^d \mathcal{N}(c_i | \mu_{i,\zeta}(D^s), \sigma_{i,\zeta}(D^s)) = \mathcal{N}(c | \mu_\zeta(D^s), \sigma_\zeta(D^s))$$

context $D^s \rightarrow p(c | D^s) \rightarrow \text{adapt } F(\cdot; c) \rightarrow \text{predict } x_t$

✗ predefined labels u — no OOD, fail without labels

✓ k-shot context D^s — generalizes to OOD, no labels

4 Framework



Infer c from context that **shares the environment**, evolve z_t with the adapted $F(\cdot; c)$, and emit x_t — one **conditional generative model**:

$$p_\Theta(x_{0:T}^q | D^s) = \int \int p_\Theta(x_{0:T}^q | c, z_0) p_\zeta(c | D^s) p(z_0) dc dz_0$$

5 What We Prove

With **sufficient environment diversity** ($\geq 2d+1$), all three latents are identifiable up to **trivial transforms** (permutation + component-wise):

$$\text{Environment } c: c = A P(\hat{c}) + b$$

$$\text{State } z_t: \hat{z}_t = C \circ P(z_t)$$

$$\text{Function } \mathcal{F}: \hat{\mathcal{F}} = h^{-1} \circ \mathcal{F} \circ k$$

FIRST identifiability of c & F under a **general** $F(z; c)$ — Meta-iLaD even **recovers the true dim(c)**.

6 Training: Meta-Learning ELBO

A Bayesian meta-learning objective with two regularizers:

$$\mathcal{L}_{\text{ELBO}} = \sum_{D_n \in \mathcal{D}} \sum_{x_{0:T}^q \in D_n} \left\{ \mathbb{E}_{q_\zeta(c | x_{0:T}^q, D_n)} [\log p_\Theta(x_{0:T}^q | c, z_0)] - \text{KL}(q_\zeta(c | x_{0:T}^q, D_n) \| p_\zeta(c | D^s)) - \text{KL}(q_\zeta(z_0 | x_{0:T}^q) \| p(z_0)) \right\}$$

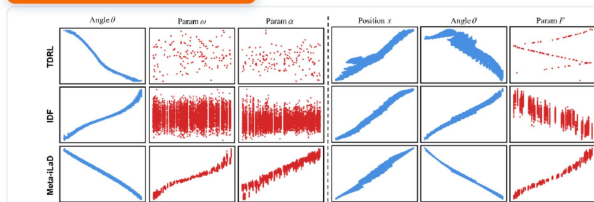
$$\mathcal{L}_{\text{TC}} = \mathbb{E}_{q_\zeta} [\log \frac{q_\zeta}{q_c}], \quad \hat{q}_c = \prod_{j=1}^d q_c(c_j), \quad \mathcal{L}_{\text{Flow}} = \mathbb{E}_{q_\zeta, q_\Theta(z_0 | x_{0:T}^q)} \left[\sum_{t=1}^T \text{KL}[q_\zeta(z_t | x_{t:T}^q) \| q_\zeta(z_t | c)] \right]$$

decorrelate c

warm-up, stabilize z

$$\mathcal{L}_{\text{Total}} = -\mathcal{L}_{\text{ELBO}} + \lambda_{\text{TC}} \mathcal{L}_{\text{TC}} + \lambda_{\text{Flow}} \mathcal{L}_{\text{Flow}}$$

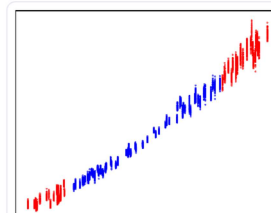
7 Identifiability Holds



Estimated vs. true latent state z_t (blue) and environment c (red). Meta-iLaD recovers **both**; label-based TDR / IDF fail to identify c .

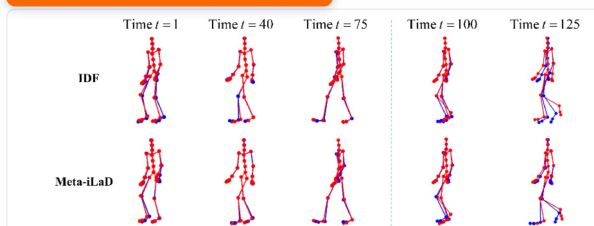
8 Forecasting & OOD

Data	Method	Pred MSE↓	MCC(c)↑	OOD MSE↓
Pend.	Meta-iLaD	0.36	0.98	0.59
	MoNODE	4.60	0.30	5.80
	IDF	12.60	0.02	n/a
Cart.	Meta-iLaD	0.27	0.99	0.49
	MoNODE	5.10	0.20	5.50
	IDF	1.50	0.16	n/a



Environment c stays identifiable **out-of-distribution** — in-range (blue) & out-of-range (red) track the truth.
n/a: IDF can't run OOD.

9 Real-World Motion Capture



Held-out CMU subjects — Meta-iLaD gives the **lowest OOD error (78.7 vs. IDF 118)**. Red = estimate, blue = truth.

